

Eindhoven University of Technology

AI-Supported Sustainability Self-Assessment for TU/e Research

ENSU – Written Assignment Stage 1

Course:	5ARF0 Ethics of AI: Theory and Practice
Group members:	[TODO: Names and student numbers]
Clients:	Philip Nickel and Tijn Borghuis
Primary coach:	Qin Tao (Tata) Zeng
Date:	[TODO: Submission date]
Word count:	[TODO: Main-text word count]

1 Introduction and Motivation (500 words)

Topic and motivation. [TODO: Define the sustainability problem in TU/e research. Explain why an AI-supported tool may help. Cite evidence.]

Objectives and terms. [TODO: State one project aim and one research question. Define the essential terms.]

Context and scope. [TODO: Specify users, research activity, decision point, setting, and non-goals.]

Values and needs. [TODO: Introduce stakeholder needs and the main values: sustainability, transparency, accountability, fairness, and human oversight.]

Methods. [TODO: Explain why Design for Values or Value Sensitive Design fits the aim. Introduce conceptual, empirical, and technical investigations with references.]

Logic: research context → stakeholder need → value → requirement → tool feature → success indicator.

2 Initial Results (1,000 words)

Preliminary literature search. [TODO: Report databases, search terms, dates, selection criteria, number of sources, and main evidence gaps.]

Table 1: Literature-search log.

Source	Type	Main finding	Need or value	Design implication
[TODO: Citation]	[TODO: Type]	[TODO: Finding]	[TODO: Need/value]	[TODO: Implication]
[TODO: Citation]	[TODO: Type]	[TODO: Finding]	[TODO: Need/value]	[TODO: Implication]

Preliminary findings. [TODO: Synthesize researcher needs, current assessment methods, AI opportunities, known risks, and evidence gaps.]

Stakeholders, needs, and values. [TODO: Identify users, affected groups, governors, experts, and their roles in the design.]

Table 2: Stakeholder and value map.

Stakeholder	Need	Value	Benefit or harm	Design role
TU/e researchers	[TODO: Need]	[TODO: Value]	[TODO: Benefit/harm]	[TODO: Role]
[TODO: Stakeholder]	[TODO: Need]	[TODO: Value]	[TODO: Benefit/harm]	[TODO: Role]
[TODO: Stakeholder]	[TODO: Need]	[TODO: Value]	[TODO: Benefit/harm]	[TODO: Role]

Tool concept and purpose. [TODO: Describe the user task, inputs, AI function, outputs, next human decision, and connection to the current TU/e self-assessment tool. State how experts can modify it.]

Requirements and constraints. [TODO: Separate functional, non-functional, ethical requirements, constraints, and non-goals.]

Table 3: Requirement traceability matrix.

ID	Stakeholder need	Value	Requirement	Tool feature	Success indicator
R1	[TODO: Need]	Sustainability	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]
R2	[TODO: Need]	Transparency	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]
R3	[TODO: Need]	Human oversight	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]
R4	[TODO: Need]	[TODO: Value]	[TODO: Requirement]	[TODO: Feature]	[TODO: Indicator]

Design strategy. [TODO: Describe the steps from evidence and stakeholder input to requirements, mock-up, expert review, evaluation, and revision.]

3 Discussion (500 words)

Ethical issues and value conflicts. [TODO: Discuss two or three project-specific conflicts. Link stakeholders, values, design choices, ethical frameworks, and remaining limitations.]

Table 4: Ethical risks and responses.

Risk	Stakeholders	Values	Response	Limitation
Privacy	[TODO: Who]	Privacy; autonomy	[TODO: Response]	[TODO: Limit]
Unclear AI advice	[TODO: Who]	Transparency; accountability	[TODO: Response]	[TODO: Limit]
Greenwashing	[TODO: Who]	Sustainability; trust	[TODO: Response]	[TODO: Limit]
Rebound effects	[TODO: Who]	Environmental impact	[TODO: Response]	[TODO: Limit]
Unfair recommendations	[TODO: Who]	Fairness	[TODO: Response]	[TODO: Limit]
Reduced human control	[TODO: Who]	Human oversight	[TODO: Response]	[TODO: Limit]

Barriers and challenges. [TODO: Discuss relevant limits in evidence, representation, data, bias, explainability, adoption, AI energy use, and project scope.]

Next steps. [TODO: State what expert feedback or prototype evaluation must test next.]

References

- Antoine Berthelot, Eddy Caron, Maxime Jay, and Laurent Lefèvre. Understanding the environmental impact of generative ai services. *Communications of the ACM*, 68(7):46–53, 2025. doi: 10.1145/3725984.
- Gry Hasselbalch and Aimee van Wynsberghe. Artificial intelligence, power, and sustainability. In Nathalie A. Smuha, editor, *The Cambridge Handbook of the Law, Ethics and Policy of Artificial Intelligence*, pages 117–130. Cambridge University Press, 2025.

A Documentation of Process

[TODO: Record meetings, research, design work, writing work, decisions, and evidence.]

Table 5: Project process log.

Date	Participants	Activity	Output or decision	Next action
[TODO: Date]	[TODO: Names]	[TODO: Activ- ity]	[TODO: Output]	[TODO: Action]
[TODO: Date]	[TODO: Names]	[TODO: Activ- ity]	[TODO: Output]	[TODO: Action]

B Project Planning

Table 6: Stage 1 project schedule.

Date	Activity	Owner	Deliverable
8 September	Fix scope and research question	[TODO: Owner]	Scope statement
9–10 September	Conduct literature search	[TODO: Owner]	Search log and sources
11 September	Map stakeholders and values	[TODO: Owner]	Stakeholder map
12–13 September	Define tool concept and flow	[TODO: Owner]	Concept sketch
14 September	Specify requirements and risks	[TODO: Owner]	Traceability tables
15 September	Complete first draft	[TODO: Owner]	Full draft
16 September	Review grading criteria	[TODO: Owner]	Revision list
17 September	Revise and proofread	[TODO: Owner]	Final draft
18 September	Submit before 23:59	[TODO: Owner]	Stage 1 submission